Soknarith (Soki) Sem

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Summary

Materials scientist with training in chemical engineering and applied mathematics, specializing in statistical thermodynamics, transport phenomena, and atomistic modeling. Experienced in first-principles quantum mechanics simulations (VASP & GPAW), classical molecular dynamics simulations (LAMMPS), and physics-based modeling to quantify defect energetics, transport behavior, and temperature-dependent properties.

Work Experience

Research Assistant & Graduate Teaching Assistant — **May 2024 to Present**
Worcester Polytechnic Institute, Worcester, MA

- Developed physics-based thermodynamic and transport models for defect-mediated ionic conductivity in Ruddlesden–Popper (RP) phase oxide perovskites
- Computed temperature-dependent free energies and defect energetics using first-principle (DFT) simulations
- Analyzed structure–property–process relationships governing ionic transport across temperature regimes
- Assisted instruction in Applied Analytical Methods (ODEs/PDEs) and Metal Additive Manufacturing

Chemical Process Engineer — **May 2023 to May 2024**
Knowles Precision Devices, New Bedford, MA

- Led R & D-to-production transition for high-voltage capacitor dielectric fluids
- Optimized manufacturing layout and process flow, reducing production costs by 15%
- Authored SOPs and technical documentation to support scale-up and cross-team transfer

R & D Technician — **Mar 2023 to May 2023**
Factorial Energy, Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments
- Reduced material waste by 20% through inventory system redesign

Purification Engineer — **Jul 2022 to Mar 2023**
Eurofins, Framingham, MA

- Purified therapeutic proteins using AKTA Avant systems (>95% purity)
- Extended TFF membrane lifetime by 30% through process optimization

Projects

Defect Chemistry of RP Phase $(La,Nd,Pr)_2NiO_{4+\delta}$ Study — **2024 to 2026**

- Performed DFT calculations on certain oxide RP phase perovskites.
- Calculated the defect formation energy, density of states (DOS), phonon density of states (PhDOS), elastic tensor, dielectric tensor, defect free energy and chemical potential phase diagram.

Bio-based Asphalt Molecular Dynamics Study — **2020 to 2022**

- Performed classical molecular dynamics simulations to characterize temperature-dependent transport behavior
- Extracted viscoelastic properties across multiple temperature regimes to inform material performance

Carbon Nanotube Purification Study — **2019 to 2020**

- Designed and implemented purification protocols achieving 95% nanotube homogeneity

Education

M.S. Materials Science & Engineering — **GPA: 3.50**
Worcester Polytechnic Institute, Worcester, MA — **2024 to 2026**

B.S. Chemical Engineering & B.S. Applied Mathematics — **GPA: 3.16**
University of Rhode Island, Kingston, RI — **2019 to 2022**
Minors: Chemistry, Physics

Technical Skills

Atomistic Simulation:	Density Functional Theory (VASP & GPAW), Molecular Dynamics (LAMMPS)
Scientific Computing:	Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX
Theory & Modeling:	Thermodynamics, Transport Phenomena, Defect Chemistry
Laboratory:	Glovebox operation, HPLC, AKTA FPLC, TFF, SEM/XRD, DSC/TGA
Process Engineering:	Manufacturing optimization, SOP development, statistical process controls